B.Comp. Dissertation

AI in financial data analytics and trading

Budgeted event screening for structural arbitrage

By

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Project ID: H398200

Advisor: Prof Yao Lu

Deliverables: report (one volume); FYP Python package (28 tests); research.ipynb; artifacts (CSV tables and 50 figures).

Abstract

This dissertation investigates whether hourly event-level order-book features can allocate limited monitoring capacity to structural arbitrage in prediction markets. A frozen Dome cohort contains 46 negRisk and five ladder events. Coverage-aware labels distinguish observable opportunities from missing evidence, while a separate economic replay isolates execution assumptions from learning targets. Thirteen weekly expanding-history fits compare rules, logistic regression, random forests, XGBoost, two-stage value models, TabM and rank ensembles. At a primary budget of three events, the fixed book-feature forest captures 94.71% of known positive hours and 99.53% of peak quoted value, versus 73.54% and 60.76% for the quoted-gap rule. Under a 100 ms, uncapped-capital replay, their known simulated P&L is $5,873.32 and $4,370.05 respectively. However, the December Fed event supplies 83.15% of the quoted-value denominator; excluding it removes the forest's economic advantage. Metadata, tuning and ensembling provide mixed gains, and TabM does not lead. These retrospective results support budgeted opportunity screening within the observed cohort while exposing substantial concentration, coverage and execution limitations. The submission provides raw inputs, source, tests, a reproducible notebook and indexed artifacts.

Subject descriptors: Machine learning; Simulation evaluation; Financial markets.

Keywords: prediction markets; structural arbitrage; event ranking; order-book replay; tabular learning.

Implementation: Python 3.12; pandas; scikit-learn; XGBoost; PyTorch/TabM; local parquet data. CUDA acceleration is optional.

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1 Introduction

1.1 Motivation

Prediction markets turn uncertain future outcomes into tradable contracts. Their prices provide information about participants' beliefs, but related contracts also impose logical constraints on one another. A complete collection of mutually exclusive outcomes should not collectively cost less than its guaranteed payoff. Likewise, a narrower condition cannot occur without a compatible broader condition. Violations can produce structural arbitrage baskets without requiring a forecast of the eventual winner.

The practical difficulty considered here is deciding where to look. Continuously processing every market's depth, checking relationships and maintaining order state consumes resources. This dissertation represents that constraint as a fixed event-monitoring budget. At scheduled times, a screening policy observes all eligible events and chooses a small subset for intensive monitoring during the following period. It must allocate resources before knowing the future opportunity path.

The budget is an experimental abstraction. The study does not measure exchange subscription limits or demonstrate that monitoring the entire contemporary platform is impossible for a particular trading firm. Instead, it asks whether limited monitoring can preserve most of the observable opportunity available in a fixed historical cohort. A useful screen could support systems where relationship checking, execution infrastructure or risk supervision is costly.

1.2 Research question and objectives

The central question is: can features computed from an event's scheduled order-book state predict which events will offer depth-supported structural arbitrage during the next hour, sufficiently well to allocate only three monitoring slots? Five slots provide a secondary budget comparison.

Four objectives make the question testable. First, construct event-level examples whose features are available at the scan and whose future labels distinguish negative evidence from missing evidence. Second, compare simple observable rules with linear, tree-based and recent neural tabular models, including bounded tuning and rank ensembles. Third, identify whether additional metadata and modelling complexity improve capture. Fourth, evaluate the same monitoring decisions under separate execution assumptions, including latency and incomplete baskets.

The distinction between opportunity and execution is central. A model should learn where useful opportunities tend to occur, rather than learn losses created by one particular latency or unwind setting. Conversely, strong opportunity capture does not establish economic

profitability. Both layers are needed, and neither should substitute for the other.

1.3 Contributions and scope

The contribution is an implemented research workflow for budgeted event selection, rather than a new general learning architecture. It combines coverage-aware occurrence and value labels, comparable chronological models, and persistent economic replay. The submission includes the relevant raw books, frozen contract membership, prepared examples, trained weekly models, 28 focused tests, an executable notebook, CSV tables and a catalog of 50 figures. Saved checkpoints can be verified or fully reconstructed from raw data.

The empirical scope is a selected Dome cohort: 46 negRisk complete-set events and five ladder events. Two buy-side basket structures are implemented. The study does not cover arbitrary natural-language relationships, all sell-side or complement constructions, or the entire Polymarket universe. Model choice is a retrospective comparison of chronological predictions; it is not described as a pristine, untouched final-test selection. These boundaries determine the strength of the conclusions.

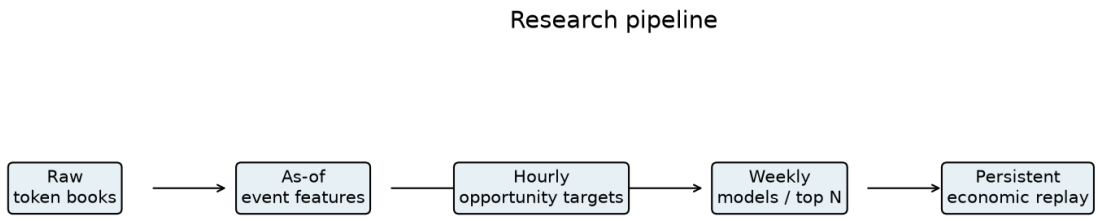


Figure 1. Research pipeline. Opportunity targets and tuning are independent of execution P&L.

2 Background and related work

2.1 Forecasting and structural consistency

Wolfers and Zitzewitz (2004) discuss prediction markets as mechanisms that aggregate dispersed information about uncertain outcomes. Their emphasis is the quality and interpretation of market-generated forecasts. This motivates studying these markets but does not establish that every relationship between contracts is continuously arbitrage-free. Forecast accuracy and structural consistency are distinct: an event can be difficult to predict while its contracts still admit a guaranteed-payoff basket.

An event in this study is one ranked object, such as a particular Fed decision. It can contain many binary markets. A token is one YES or NO outcome share and has its own order book. A basket is an audited set of tokens with a known payoff lower bound. A candidate is an event eligible for a scan; the term does not mean an opportunity already exists. A signal is an observed interval in which a basket passes the opportunity test. An attempted order, a completed basket and a profitable executed trade are separate observations.

2.2 Empirical arbitrage research

Saguillo, Ghafouri, Kiffer and Suarez-Tangil (2025) investigate arbitrage on Polymarket, distinguish within-market rebalancing from relationships across markets, and address the cost of searching related conditions using heuristic reduction. Their evidence motivates asking whether mispricing exists rather than assuming a perfectly consistent platform. The task here differs: it predicts which events deserve future monitoring under an explicit slot budget. The cited market-wide profit estimate is neither a label nor a profitability benchmark for this dataset. Historical depth coverage, basket definitions and execution assumptions must be established independently for the present study.

This distinction also limits the role of semantic discovery. A market title suggesting a related subject is insufficient to guarantee compatible settlement rules. The implemented cohort therefore uses frozen audited relationships. Membership is an upstream assumption, whereas the learner ranks events after membership has been fixed. General automatic relationship discovery remains outside the evaluated strategy.

2.3 Tabular learning methods

Breiman (2001) introduces random forests as ensembles of randomized tree predictors. For this application, their ability to express nonlinear feature interactions provides a useful comparison with a regularized linear score. The original method does not resolve temporal leakage, class imbalance or resource-constrained evaluation; those require application-specific choices.

Chen and Guestrin (2016) present XGBoost as a scalable tree-boosting system. Here it provides a sequential tree ensemble alongside the forest's averaging approach. Its inclusion tests a different tabular inductive bias, rather than presuming that a stronger general benchmark result implies better event ranking. All compared families receive the same chronological populations and bounded selection objective.

Grinsztajn, Oyallon and Varoquaux (2022) compare neural and tree methods across tabular benchmarks and examine why trees can remain competitive on medium-sized tables. Their findings justify retaining strong classical controls. However, general tabular benchmarks are not evidence about rare, temporally dependent prediction-market opportunities. A comparison in the present setting is still necessary.

Gorishniy, Kotelnikov and Babenko (2025) introduce TabM, an ICLR 2025 method using parameter-efficient ensembling within a tabular neural model. It supplies a recent neural comparator rather than a generic default multilayer network. This study uses the official implementation and preserves member-wise training losses and prediction averaging. Its smaller bounded search and three-seed two-stage adaptation are application choices, not a replication of the paper's full benchmark programme. Success on broader benchmarks does not guarantee superiority on this small, concentrated cohort.

2.4 Evaluation and interpretation

Bailey, Borwein, López de Prado and Zhu (2017) discuss overfitting when backtests are used to select strategies and propose a framework for assessing it. The present study does not implement their probability-of-backtest-overfitting estimator. Its practical response is to separate earlier-only tuning from forward prediction, preserve simple controls, report concentration and qualify retrospective policy selection. Chronological prediction alone does not remove all uncertainty created by a selected cohort and multiple policy comparisons.

The resulting research gap is a specific resource-allocation problem. Evidence that arbitrage has occurred does not show that an hourly screen can anticipate it. A high-performing generic learner does not show that its selected events will produce superior executed returns. The experiments below connect those questions while retaining their separate targets and assumptions.

3 Problem formulation

3.1 Scheduled ranking

Let $\mathcal{C}(t)$ be the eligible event set at scheduled scan t. For each event e, the screen constructs $\mathbf{x}(e, t)$ using only current and earlier recorded states. A policy assigns score $s(e, t)$ and selects the highest-scoring $\min(N, |\mathcal{C}(t)|)$ events. Ties use the event identifier deterministically. Future labels do not filter $\mathcal{C}(t)$ or the selected rows.

The selected events are monitored over the half-open interval $[t, t + H)$, where H is one hour. An opportunity at the scan counts in that interval; a new opportunity exactly at $t + H$ belongs to the next window. The implementation makes cadence and horizon configurable but requires them to match, preventing overlapping allocations from silently exceeding the monitoring budget. The reported models and results use only the one-hour setting.

When $|\mathcal{C}(t)| \leq N$, every candidate is selected. These hours cannot distinguish exclusion quality. The study therefore reports candidate competition, competitive-hour recall and the larger-budget comparison alongside pooled capture. Equal slot counts also need not imply equal processing cost: events can require very different numbers of token books.

3.2 Guaranteed-payoff baskets

For an audited mutually exclusive, exhaustive negRisk event, buy one YES share of every member. Exactly one member resolves YES, so one complete basket pays \$1. For q complete shares, the conservative payoff is q dollars. All members are required; an unavailable leg cannot simply be dropped.

For a ladder, let A be a broader condition and B a narrower condition such that $B \Rightarrow A$ under compatible resolution rules. Buying $\mathrm{YES}(A)$ and $\mathrm{NO}(B)$ pays one dollar if neither occurs, two dollars if A occurs without B, and one dollar if both occur. The impossible state B without A is excluded by the audited implication. The simulator uses the guaranteed one-dollar lower bound and does not add a speculative payment for the intermediate outcome.

For a basket with m members, let $C_j(q, t)$ be the integrated ask cost of buying q shares of member j at time t. The quoted lower-bound margin is

$$G(q, t) = q - \sum_{j=1}^{m} C_j(q, t). \tag{1}$$

Reading only best asks is insufficient because depth can be exhausted before q shares are filled. A qualifying quantity must meet exchange minimums, integer sizing and tick constraints. The study requires $q \geq 5$ and places no artificial cash or share cap.

The marginal limit $L_j(q, t)$ is the worst ask price required for leg j. Entry qualification additionally requires

$$q - q \sum_{j=1}^{m} L_j(q, t) > 0. \tag{2}$$

This guards against submitting independent limit orders whose worst supported combined cost exceeds the guaranteed payoff, even if their integrated displayed cost looks favourable. There is no fixed \$0.50 profit deduction or minimum-dollar threshold.

3.3 Occurrence, value and missing evidence

The occurrence target $y(e, t)$ is one when any active basket qualifies during the following hour and zero when none does. Both classes require the active window to be fully decidable from the recorded states. If a required interval is unknown, y remains missing even when a positive gap was witnessed elsewhere in that hour. This symmetric requirement prevents partially observed positives from being compared with artificially complete negatives.

The value target $v(e, t)$ is the largest qualifying one-basket dollar margin observed during the window. It is zero for a known negative and missing for an unknown outcome. Repeated snapshots do not add value. Multiple ladder pairs are not summed because their depth can overlap. Consequently, summing v over event-hours is an evaluation weight, not a statement of cumulative tradable profit.

A secondary diagnostic distinguishes opportunities already visible at the scan from newly observed onsets afterwards. The first layer is not restricted to classifying whether a positive gap is visible now. It is evaluated on the future monitoring period, including the possibility of new opportunities after the selection decision.

3.4 Metrics and references

Positive-hour recall is the number of selected known positive event-hours divided by all known positive event-hours. Quoted-value capture is the selected known v total divided by the full known v total. The primary model-comparison criterion is pooled value capture at N=3, with occurrence capture and average precision used to resolve ties. Precision among known selected rows, average precision, unknown-slot share and token-monitoring burden provide additional perspectives.

Random selection has an exact expected inclusion probability $\min(N, m)/m$ in a scan with m candidates. Its expected positive and value capture can therefore be calculated without Monte Carlo noise. The economic random baseline instead uses 20 explicit fixed-seed policies, because trading path dependence prevents simple linear expectation of P&L. Monitor-all supplies a full-resource reference. Neither a zero-positive week nor a zero-value week defines

a recall denominator; those metrics remain undefined rather than becoming zero.

4 Data and example construction

4.1 Source, cohort and coverage

The source is saved Dome token-level L2 depth with frozen Polymarket metadata. The supplied raw cohort contains 994 token histories, approximately 1.01 GB. The calendar extends from 14 October 2025 to 3 February 2026 at 08:00 UTC. Its 112 days and eight hours are not continuous complete coverage for every event. Only the two audited basket types enter the study.

There are 51 supported events: 46 negRisk and five ladders. The complete prepared table has 29,308 eligible event-hours, 23,452 known targets and 470 positive targets. The forward interval contains 44 distinct events. Some supported events contribute earlier training history but no usable forward scans. The relatively small ladder sample is a limitation of the available complete histories and audited relationships, not evidence that ladder arbitrage is rare throughout the market.

Table 1. Forward ranking population.

Measure	Value
Scheduled hours	2,024
Event hours	19,766
Events	44
Known	15,787
Unknown	3,979
Positive	359
Positive events	12
Median candidates	9.00
Maximum candidates	19
Competitive hours n3	1,887
Total peak value	8,838.31

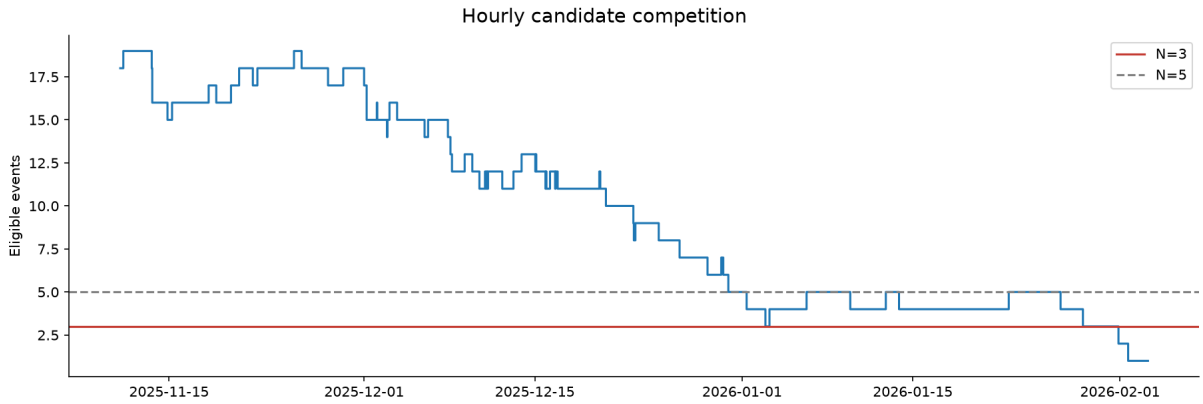


Figure 2. Hourly candidate competition. Candidate counts use the complete eligible pool including unknown labels.

4.2 Raw depth and as-of alignment

Each raw row includes source snapshot time, provider index time, ask and bid level lists, minimum order size and tick size. Prices and sizes are checked for finite valid values. Empty depth is distinguishable from missing or malformed depth. Crossed books do not provide trusted entry evidence.

At a scheduled scan, each active token uses its latest source-time row at or before the scan. Equal source timestamps retain the last provider-indexed version with stable ordering. Tokens may have different last-update times; their book ages quantify that difference. No future book seeds an earlier state. During replay, each token is additionally bounded by its own audited recording end.

Using source time approximates observation through an idealised feed subject to the declared total execution delay. Provider receive-time delays are not reconstructed. A 100 ms scenario therefore does not verify that the historical collector delivered each source update within that delay. Removing a freshness cutoff likewise avoids arbitrary exclusion but does not prove that all carried states remained available at the exchange.

4.3 Eligibility and full-window labels

An event enters a scan when its active basket is valid, all required members have been seen and the scan lies within the offline event observation extent. This is separate from label coverage. It permits an event with a usable current snapshot but an unknown future interval to compete fairly for a slot. Training subsequently excludes that unknown target; selection does not.

Coverage is evaluated over active basket intervals. Normally every required leg must be observed. An observed blocker, such as empty asks on one required leg, can establish that a basket cannot be bought even if another leg is missing. Otherwise missing member evidence

makes the interval unknown. Both occurrence and value require full-window decidability under this rule.

Raw reconstruction preserves an audited preprocessing boundary at 8 January 2026 00:00 UTC. Earlier coverage ends at the last available state within its partition; later coverage uses audited per-token endpoints and bounded exit context. This boundary is independent of training and forward-evaluation origins. Preserving it avoids silently turning previously unknown edge windows into known labels. Scheduled feature lags remain continuous where the preceding scan is available.

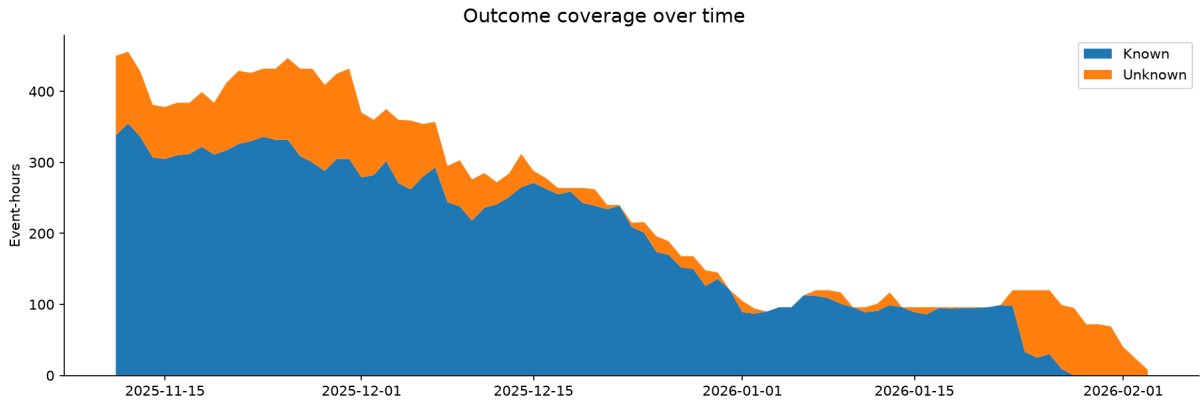


Figure 3. Outcome coverage over time. Unknown intervals do not become negative examples.

4.4 Features and metadata

There are 11 current book summaries: active baskets, active tokens, seen-member fraction, clean-member fraction, mean spread, maximum spread, total ask depth, maximum book age, best quoted dollar margin, size-feasible basket fraction and positive quoted-margin basket fraction. Each has one consecutive-scan lag and one difference from that lag, giving 33 book features. Lag construction precedes snapshot eligibility filtering, and a gap in the scheduled grid yields a missing lag rather than an invented observation.

The quoted-gap feature maximises integrated quoted margin over supported integer sizes and active baskets. It may be negative. It omits the positive worst-limit-margin requirement used by opportunity qualification and execution; it is a descriptive ranking signal, not an instruction to submit every basket that it scores. Missing quoted margins receive the rule's lowest score.

Three optional metadata features describe event type, fixed broad topic and days since the earliest frozen member's creation. Creation age is a proxy for event maturity, not a verified historical listing date. Negative or missing ages stay missing. Broad topic uses fixed title-based categories; event IDs, winners, realised status, future volume and execution returns are excluded. Frozen topic text and membership retain a retrospective selection limitation.

Metadata ablation compares fixed book-only and fixed combined models with identical family settings. It therefore isolates adding the three metadata features more cleanly than comparing separately tuned models. Feature correlations are descriptive; they are not causal explanations or proof of independent predictive contribution. The current seen-member fraction is mechanically constant among eligible scans, which illustrates why feature availability alone should not be confused with useful predictive variation.

5 Prediction methodology

5.1 Rules and model families

Three rules establish low-complexity references. Quoted gap ranks the best integrated basket dollar margin at the scan. Depth ranks total active ask shares. Freshness ranks the negative maximum member book age, preferring events whose oldest active state is more recent. None of the rules needs the eventual label, and a negative quoted gap can still receive a relative rank.

Logistic regression models a regularized linear boundary for opportunity occurrence. Its simplicity provides a useful test of whether current feature combinations already explain most ranking quality. Random forest models nonlinear interactions through an ensemble of trees. XGBoost supplies a regularized boosted-tree alternative. Each family has fixed book-only, fixed combined-feature and tuned combined-feature policies.

Two-stage models estimate expected opportunity value from an occurrence probability and a conditional positive-value estimate:

$$\hat{v}(\mathbf{x}) = \widehat{\mathbb{P}}(y = 1 \mid \mathbf{x})\, \widehat{\mathbb{E}}[v \mid y = 1, \mathbf{x}]. \tag{3}$$

The probability gate uses natural prevalence, rather than class weighting intended to rebalance classification. The amount model trains on positive quoted values only, with nonnegative predictions. A fixed forest hurdle is a reference; TabM uses the same two-stage decomposition. There is no hard probability threshold before ranking.

Three ensembles average within-scan percentile ranks: the three tuned classifiers; tuned logistic, tuned forest and TabM; or the two value models. Percentiles permit comparison of scores with different scales. Weights are equal and fixed. Combining models is an empirical comparison, not an assumption that ensembles must improve upon their strongest component.

5.2 Preprocessing and bounded tuning

Classical numeric preprocessing uses training-fitted median imputation with missing indicators, a signed log transform and standardisation. Metadata categories use training-fitted one-hot encoding with unknown categories ignored. The same pipeline interface keeps feature order and preprocessing inside each fitted artifact.

Each classical family evaluates eight configurations. Logistic settings vary regularisation strength, L1/L2 penalty and class weighting. Forest settings vary depth, leaf size and feature subsampling with 200 trees. XGBoost settings vary tree count, depth, learning rate, child weight, subsampling and regularisation. The exact grids, search results and selected settings are exported, rather than replaced by a claim of exhaustive tuning.

TabM uses 32 implicit members per stage and four configurations selected from the declared architecture grid. Numeric preprocessing is fitted on training data; continuous variables use quantile transformation and categories use one-hot encoding. Member losses are computed separately, and inference averages member probabilities or amount predictions. Batch size is 1,024, the maximum is 60 epochs and early-stopping patience is 10. Inner tuning uses one fixed seed. Weekly refits average three seed-specific gate-times-amount products. Refit epochs are the ceiling of the median selected inner best epoch, carried forward until the next tuning origin.

5.3 Expanding chronological evaluation

The initial 28 calendar days supply training history. Forward evaluation runs from 11 November 2025 to 3 February 2026 at 08:00 UTC: 2,024 scheduled hours, or 84 days and eight hours. There are 13 weekly refits, with an eight-hour final block. A training example enters only if its known target window finishes strictly before the fitting origin. Later observations can enter subsequent fits after that requirement is satisfied.

Tuning occurs at the first origin and every fourth weekly origin when at least seven forward days remain: 11 November, 9 December and 6 January. Two inner chronological blocks lie entirely in the earlier history. Their full target horizon is purged before each inner fit. Configurations maximise pooled captured quoted value at N=3, then captured positive hours, mean inner average precision and declaration order. No subsequent weekly outcomes determine that origin's hyperparameters, early stopping or ensemble weights.

There are 168 inner candidate evaluations. The saved weekly audit contains 199 distinct fitted artifacts or stages; identical selected and fixed classifier settings can share an artifact. They are not counted as independent supporting evidence. The known training population grows from 7,652 rows initially to 23,452 at the final fit. The final origin does not imply recent useful data: the last known training scan is on 27 January because later coverage is missing.

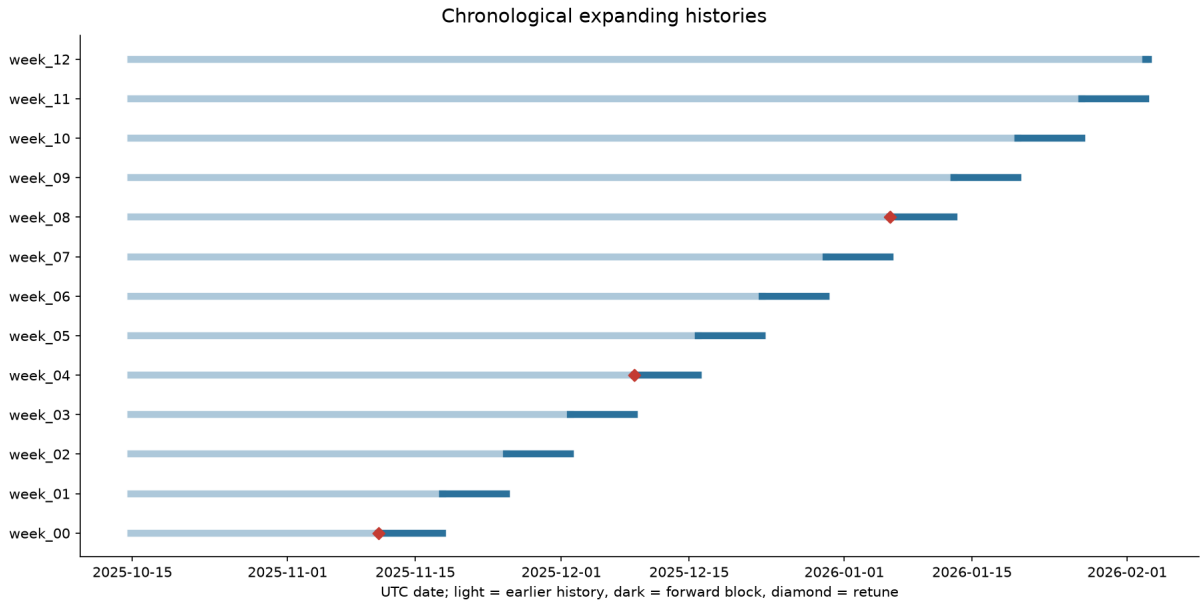


Figure 4. Chronological expanding histories. Full target windows are purged before each origin. Final block is eight hours.

5.4 Model choice and uncertainty

The final recommended policy is selected by the pooled opportunity criterion from the saved chronological comparisons. This retrospective policy choice does not alter the already generated weekly scores. It also does not establish untouched-test generalisation. Economic P&L is reported afterwards and does not replace the opportunity criterion merely because a different model has higher simulated returns.

Paired seven-day block resampling compares policies on the same scheduled periods. Ten thousand draws preserve within-block dependence more faithfully than resampling individual event-hours. There are only 13 blocks, including a short last block. Intervals condition on the fitted models and observed cohort; they do not include refitting, universe selection, missing outcomes or a multiple-comparison correction. Event-exclusion analysis therefore remains essential alongside intervals.

6 Economic replay

6.1 Separate execution layer

The economic layer fixes the first-layer selections and varies execution assumptions. The primary scenario uses 100 ms total signal-to-order-arrival delay, immediate submission, uncapped capital and shares, and available-depth sizing. There is no imposed random failure probability, minimum signal duration or quote-age ceiling. Entry constraints still enforce valid depth, exchange minimums, ticks and positive worst-limit margin.

The frozen cohort uses zero exchange fees and zero extra per-attempt charges. Saved market flags support this scenario; they are not a timestamped historical fee-change ledger. The results should not be extrapolated to fee-bearing contracts without rebuilding qualification, sizing and replay costs.

100 ms represents an assumed observation and execution system, not a measured live fill delay. Other deterministic scenarios use 0, 50, 250 and 500 ms. The zero-delay scenario is an optimistic recorded-state benchmark, not a verified upper bound on the exchange's available profit. Every scenario retains the same event selections and uses its own persistent ledger.

6.2 Submission, arrival and partial baskets

When a selected event produces a new qualifying signal, the scheduler sizes the best supported basket using the decision-time books and freezes each leg's price limit. If several pairs start together, decision-time quoted margin determines their order, with a deterministic pair-ID tie break. Future arrival outcomes do not decide which basket to submit.

At arrival, each leg is assumed to fill in full if the recorded ask depth can support its entire requested quantity within its frozen limit. Otherwise that leg does not fill. This is independent per-leg FOK support, not an atomic multi-market transaction. Missing arrival evidence remains unknown rather than being interpreted as a failed fill.

A completed basket records conservative payoff minus actual integrated entry cost. If no legs fill, P&L is zero in this no-attempt-charge scenario. If some legs fill, the simulator tries to sell them at recorded bid depth one second after arrival. Their proceeds minus entry cost produce the partial result. Missing or insufficient exit evidence leaves that result unknown. For example, ten shares bought at $0.40 and unwound at $0.30 lose $1; there is no synthetic percentage failure deduction.

The order is attempted when the signal becomes visible. Its eventual duration is not used as an immediate-entry filter. Short-lived opportunities can thus be positive first-layer labels while producing no fill or a partial loss after latency. That difference is an intended measurement of

execution sensitivity.

6.3 Persistent liquidity and accounting

Each event scheduler permits one attempt in flight at a time. Shared-token ask versions remain reserved so that different ladder pairs or later scans cannot spend the same unchanged displayed liquidity repeatedly. Model refits and weekly boundaries do not reset these reservations. The supplied events have no token overlap across separate event schedulers; another universe with cross-event overlap would require a global shared ledger.

Complete-basket margins are recognised at completion and partial outcomes at unwind. Recognition is distinct from receiving settlement cash. Known P&L sums exclude missing outcomes and separately count them. Drawdown starts from a zero balance and uses the resulting accounting-time increments.

Funding is unlimited in the primary comparison. A separate no-settlement scenario accumulates purchase cash without releasing completed-basket proceeds during the study, while allowing known partial-unwind proceeds and conservative unknown reservations. It estimates a funding requirement under that scenario, not deployed capital or an ROI denominator. Monitor-all uses more monitoring resources; path-dependent order competition prevents treating it as a formal profit upper bound.

7 Results

7.1 Population and monitoring competition

The forward population contains 19,766 event-hours from 44 events. There are 15,787 known outcomes, including 359 positive hours from 12 distinct events, and 3,979 unknown outcomes. Median concurrent pool size is nine, the maximum is 19, and 1,887 scheduled hours have more than three candidates. Thus the primary budget usually requires exclusion, although competition and label coverage vary materially over time.

At N=3, each deterministic policy selects 5,973 event-hours. The positive-hour random expectation is 22.23%; expected quoted-value capture is 18.93%. These are conditional expectations for the observed candidate sets, not estimates of market-wide opportunity prevalence. Known targets sum to $8,838.31 of hourly peak quoted value. That number is a weighting denominator rather than a cumulative executable profit pool.

7.2 Primary opportunity comparison

Table 2. Opportunity capture at N=3.

Policy	Positive hours	Recall %	Value %
Forest: book	340	94.71	99.53
Forest + metadata	333	92.76	99.41
Logistic + metadata	331	92.20	99.20
Classifier ensemble	334	93.04	99.16
Tuned logistic	334	93.04	99.06
TabM two-stage	316	88.02	99.04
Logistic: book	329	91.64	98.81
XGBoost + metadata	321	89.42	98.75
Value ensemble	315	87.74	98.58
Forest two-stage	315	87.74	98.57
Tuned XGBoost	321	89.42	98.40
Logistic / forest / TabM	323	89.97	98.37
Tuned forest	328	91.36	98.02

Policy	Positive hours	Recall %	Value %
Freshness	261	72.70	95.21
Depth	182	50.70	91.92
XGBoost: book	323	89.97	82.98
Quoted gap	264	73.54	60.76

The fixed book-feature random forest leads pooled value capture at N=3: 99.53%, with 340 of 359 known positive hours captured, or 94.71% recall. Quoted gap captures 60.76% of value and 264 positives, or 73.54% recall. The forest also captures 205 of 223 known newly arising opportunity hours that had no opportunity at the scan, showing that the comparison includes more than detecting currently positive quotes.

The quoted-gap rule's disadvantage is consistent with the decision timing. It uses a single current dollar-margin statistic, whereas the label concerns the next hour and includes later changes. A presently negative event may subsequently offer valuable depth, and a presently attractive gap need not remain useful throughout monitoring. Current depth, spreads, ages and their changes can therefore improve ranking without estimating the final event outcome. This is an interpretation of the comparison, not a causal attribution to a specific feature.

The primary forest's known-selected precision is only 6.79%. A rare-event screen can nevertheless have high recall when its purpose is allocating monitoring, rather than issuing a trade on every selected hour. About 16.16% of its selected event-hours have unknown targets. Treating these as negatives would produce a different, unjustified evaluation.

7.3 Metadata, tuning, ensembles and budget

Table 3. Matched metadata effects at N=3, percentage points.

Fixed model	Recall change (pp)	Value change (pp)
logistic	+0.56	+0.39
random forest	-1.95	-0.12
xgboost	-0.56	+15.76

Adding metadata to fixed logistic regression increases value capture by 0.39 percentage points. The fixed forest decreases by 0.12 points, while fixed XGBoost improves by 15.76 points. This mixed outcome does not support requiring metadata for every family. The large XGBoost difference is especially useful as a reminder that feature effects depend on the learner, rather than being a universal ranking of feature usefulness.

Tuned models do not consistently improve upon fixed controls. TabM captures 99.04% of quoted value and 88.02% of positive hours at N=3, but does not lead. The classifier ensemble reaches 99.16% value capture; this is strong relative to quoted gap but still below the fixed book forest. Extra architecture, search and ensemble complexity therefore has limited incremental value in this cohort. A recent model is a meaningful comparator even when it loses.

At N=5, the forest captures 98.05% of positive hours and 99.99% of value. Quoted gap reaches 93.59% and 98.45%, respectively. The larger monitoring budget narrows the gap, as expected when more events are selected. This supports reporting N=3 as the principal resource-constrained comparison and N=5 as a secondary sensitivity, rather than presenting the easier budget alone.

7.4 Economic comparison at 100 ms

Table 4. Economic comparison at N=3 and 100 ms.

Policy	Known P&L $	Attempts	Complete
Forest: book	5,873.32	453	436
Forest + metadata	5,873.60	456	439
Logistic + metadata	5,892.81	434	416
Classifier ensemble	5,902.50	462	445
Tuned logistic	5,896.85	444	426
TabM two-stage	5,857.99	431	413
Logistic: book	5,878.47	438	420
XGBoost + metadata	5,915.38	449	433
Value ensemble	5,834.01	428	410
Forest two-stage	5,921.18	454	439
Tuned XGBoost	5,886.61	456	440
Logistic / forest / TabM	5,874.54	437	419
Tuned forest	5,881.06	466	449
Freshness	5,778.40	400	384
Depth	5,745.85	285	280

Policy	Known P&L $	Attempts	Complete
XGBoost: book	5,187.45	454	437
Quoted gap	4,370.05	368	355
Monitor all	5,944.97	499	481
Random mean (20 seeds)	1,123.15	124.6	119.15

The opportunity-selected forest produces $5,873.32 known simulated P&L from 453 attempts, including 436 completed baskets and 17 partial baskets. Completed margin contributes $6,290.08; partial results contribute −$416.76. Quoted gap produces $4,370.05. Monitor-all produces $5,944.97, and the 20-seed random mean is $1,123.15. None of the primary forest's attempted execution outcomes is unknown in this replay; this does not eliminate the separate unknown opportunity windows in the candidate population.

The forest's maximum known-P&L drawdown is $206.15. Its largest order is 73,067 shares. Its no-settlement funding scenario requires $1,426,124.21, illustrating why a few thousand dollars of margin must not be interpreted as a high return on a small account. Uncapped available-depth sizing permits large positions even when per-share margins are small.

The highest model P&L belongs to the forest two-stage policy at $5,921.18, not the opportunity-selected forest. Most learned policies are economically close, and freshness and depth also obtain returns relatively near the learned group. The finding is therefore stronger against quoted gap than it is for one complex model over every plausible simple screen. Economic differences did not retroactively choose the first-layer objective.

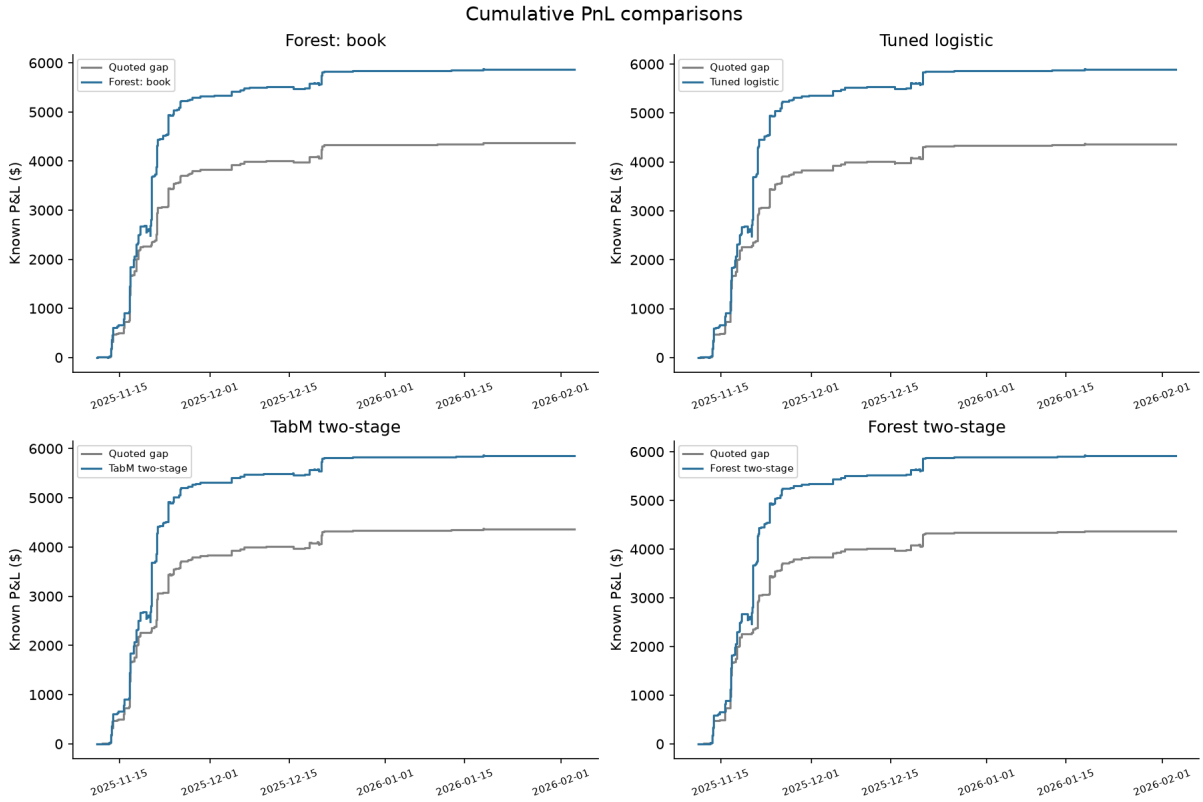


Figure 5. Cumulative PnL comparisons. N=3, 100 ms; separate panels and a common quoted-gap reference avoid crowded lines.

7.5 Latency and incomplete execution

Table 5. Selected forest: fixed selections across latency scenarios.

Delay ms	Known P&L $	Partial P&L $
0	6,471.72	0.00
50	6,345.97	-105.11
100	5,873.32	-416.76
250	5,132.97	-1,014.67
500	4,855.08	-1,190.38

The selected forest's P&L declines from $6,471.72 at zero delay to $6,345.97 at 50 ms, $5,873.32 at 100 ms, $5,132.97 at 250 ms and $4,855.08 at 500 ms. The monitoring decisions are unchanged. The difference arises from arrival-time depth support and subsequent partial baskets, not from retraining a target on the execution setting.

The zero-delay replay has no partial loss for this policy. At 500 ms, partial results contribute −$1,190.38. This illustrates why assuming simultaneous instantaneous fills would materially

change the economic conclusion. It also shows the value of keeping execution separate from learning: the same opportunity predictor can be studied under several operational systems.

These are scenario comparisons, not an empirical estimate of what a particular high-frequency trader can achieve. Source-time snapshot resolution, missing updates, competing traders and real exchange queue behaviour remain outside the supported fill model. Faster simulated execution should not be reported as measured infrastructure performance.

7.6 Concentration and uncertainty

The December Fed event contributes 83.15% of the quoted-value denominator. This concentration makes the impressive pooled value captures highly dependent on one event. Excluding it before reranking changes the economic comparison: the selected forest produces $730.52 versus quoted gap's $736.36. The study therefore does not establish a consistent profit advantage across independent events. The exclusion belongs in the main interpretation, not only in a supplementary file.

Table 6. December Fed exclusion: reranked N=3, 100 ms.

Policy after event exclusion	Known P&L $
Forest: book	730.52
Quoted gap	736.36

The paired weekly resampling interval for the selected forest's economic difference versus quoted gap is approximately $7.93 to $3,866.59 around the observed $1,503.27 difference. With 13 dependent blocks, multiple policies, missing evidence and substantial event concentration, this is conditional uncertainty rather than proof of a universal economic advantage. Resampling weeks cannot manufacture a larger set of independent event structures.

The final short weekly block has no known opportunity labels. Such periods remain part of the monitoring and missingness description, but cannot supply defined recall or value-capture metrics. Calendar coverage and useful labelled data should therefore be reported separately.

8 Discussion and conclusion

8.1 Answers to the research questions

Within the selected Dome cohort, hourly event features support effective budgeted monitoring. The fixed book-feature forest captures substantially more known positive hours and quoted value than the quoted-gap rule at N=3. It does so while monitoring only a subset of event-hours and without using future labels to filter the ranking pool. The larger N=5 budget reduces the incremental benefit of better ranking.

The comparison does not show that greater model complexity reliably improves this task. Fixed classical models remain strong, bounded tuning offers mixed gains, and TabM does not lead. Metadata has family-specific effects. These are useful research findings because they constrain which additional engineering is supported by the available evidence.

Economic replay confirms that opportunity capture and executed margin are related but different. Slower arrivals can erase fills or leave losing partial baskets. The opportunity winner is not the model with the highest primary simulated P&L. Most importantly, the economic advantage over quoted gap is concentrated in one event and disappears in the exclusion comparison.

8.2 Limitations

External validity is limited by frozen retrospective event selection, only five ladder events and 12 positive forward events. Repeated hours and related contracts are not independent samples. A larger row count does not by itself establish broader event diversity, and pooled dollar weighting is dominated by December Fed.

Coverage-aware labels avoid pretending missing data are negative, but they cannot reveal the opportunities that occurred in unknown intervals. Known-only metrics may differ from performance on a continuously observed exchange. The offline recording extent and frozen metadata are also not a fully historical live catalogue.

The execution model assumes source-time observability, independent per-leg FOK support from recorded depth, zero cohort fees, zero additional costs and unlimited funding. It does not model queue position, competing consumption, network congestion, exchange acknowledgements or settlement delays as a capital constraint. Partial unwinds use a fixed one-second delay. Their losses are therefore execution-scenario outcomes, not immutable characteristics of the event prediction problem.

The chronological fit and tuning safeguards prevent future target windows from entering earlier models. They do not remove retrospective policy-selection uncertainty or adjust every

comparison for multiplicity. The final model is a cohort-level research recommendation, not a demonstrated universally best Polymarket strategy.

8.3 Future work

A stronger subsequent study would prospectively freeze a broader audited universe, preserve point-in-time membership and collect continuous source and receive timestamps. It should evaluate the selected policy on independent future events and time periods, with particular attention to ladder coverage and event-level concentration.

Operational validation should measure actual observation-to-order latency, exchange responses and fill support. Fee-bearing contracts, settlement cash release and bounded capital should be separate declared scenarios. New sell-side or complement baskets require explicit payoff, inventory and order support rather than a sign change to the existing buy logic.

The modular workflow permits other nonoverlapping screening windows and monitoring budgets, but their labels and chronological comparisons must be rebuilt. Additional tuning should be justified by independent predictive evidence, rather than used as a substitute for a larger and better observed cohort.

8.4 Conclusion

This dissertation implements a two-layer study of structural-arbitrage event selection. It demonstrates strong opportunity capture under a three-event monitoring budget within the observed Dome cohort, while showing that execution latency, partial baskets, funding and event concentration materially qualify the economic result. The practical contribution is a reproducible way to measure those separate effects, with classical and recent tabular models compared on the same chronological task.

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Appendix A Reproduction and artifact guide

The standalone FYP package separates data preparation, structural rules, models, experiments, economic replay and reporting. The notebook research.ipynb orchestrates these stages. Its cached mode verifies saved data and fitted scores and regenerates analyses; its full mode reconstructs examples from raw books, retunes and refits models, and reruns persistent execution into a new run directory. Neither mode requires a parent repository or provider token.

The data and simulator guides document schemas and assumptions. The artifact table and figure indexes map every catalogued output to a CSV source or supporting dataset. The main report presents selected comparisons; the full 50-figure catalog supports additional inspection without filling the volume with redundant plots.

The 28 tests exercise payoff constraints, backward as-of reads, duplicate timestamps, recording endpoints, empty versus unknown books, integer depth sizing, uncapped quantities, tick/minimum constraints, short opportunities, incomplete labels, half-open windows, peak-value aggregation, positive-only amount fitting, unknown slot consumption, chronological purging, selection invariance, rank ensembles, primary execution settings, persistent reservations and partial/unknown execution outcomes. They test implemented safeguards, not the completeness of the original collector or the truth of real fills.

Appendix B Configuration and provenance

inputs/cohort.json records complete basket membership and validity intervals. inputs/raw_inventory.csv records every required raw history and its digest. inputs/study.json records the feature views and bounded configuration grids. Weekly bundle files record selected parameters and feature order; fit audits record the actual training populations and cutoffs. Tuning search and selected hyperparameter CSVs provide the detailed configurations supporting Chapter 5.

Source, input and output digests detect changes during reproduction. Verification recomputes weekly scores from fitted artifacts, checks temporal separation, reconstructs capture metrics and reconciles cumulative known P&L with economic summaries. These controls establish reproducibility of the supplied study, not point-in-time availability of frozen metadata or completeness of the feed.

B.Comp. Dissertation

AI in financial data analytics and trading

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